

AI Healthcare System - Complete Project Documentation

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Introduction

The **AI Healthcare System** is a comprehensive, multi-role web application designed to bridge the gap between patients, healthcare providers, and administrators using artificial intelligence. It features dedicated dashboards for Patients, Doctors, and Admins, facilitating everything from AI-driven symptom checking and diet planning to telemedicine (video consultations), secure medical record management, and administrative oversight.

Landing Page

Purpose:

To welcome users to the AI Healthcare System, highlight the platform's key features, and provide clear entry points for registration and login.

Features & Functionalities:

- Hero Section:** Engaging introduction with a call-to-action (CTA) to join the platform.
- Feature Highlights:** Briefly outlines core capabilities such as AI Symptom Checking, Secure Video Consultations, and Smart Prescriptions.
- Role-Based Navigation:** Clear links for Patient Login, Doctor Login, and Admin Login.

Workflow:

- The user arrives at the website URL.
- They explore the features and decide their user type (Patient or Doctor).
- They click on the respective Login or Register button, which redirects them to the Authentication pages.

(Screenshot: Landing Page)

[Insert Landing Page Screenshot Here]

Dashboard 1: Patient Dashboard

Purpose:

To empower patients with tools to manage their health, consult doctors, and leverage AI for preliminary health insights.

Pages & Functionalities:

- Overview Dashboard (/dashboard/patients)**
 - Purpose:* Provides a high-level summary of upcoming appointments, recent prescriptions, and quick AI insights.
 - Screenshot:* [Insert Patient Dashboard Screenshot Here]
- AI Symptom Checker (/dashboard/symptoms)**
 - Purpose:* Allows patients to input their symptoms and receive AI-driven preliminary diagnosis and recommendations.
- Health Chat (/dashboard/ai-chat)**
 - Purpose:* A conversational AI assistant that answers general health queries in real-time.
- Diabetes Diet Planner (/dashboard/diabetes-diet)**
 - Purpose:* Generates personalized diet plans based on the patient's vitals, weight, and specific conditions using AI.
- BMI Analysis (/dashboard/bmi)**
 - Purpose:* Tracks the patient's Body Mass Index over time with visual charts.
- Medical Reports (/dashboard/reports) & Upload Reports (/dashboard/upload-reports)**
 - Purpose:* A secure vault where patients can upload, view, and download their medical files (PDFs, images).
- Appointments & Emergency (/dashboard/appointments, /dashboard/emergency)**
 - Purpose:* Book new consultations (video/in-person) and access quick emergency contacts or immediate telemedicine rooms.
- Billing & Payments (/dashboard/billing, /dashboard/payment-history)**
 - Purpose:* Manage invoices for consultations and view past transaction history.

Dashboard 2: Doctor Dashboard

Purpose:

To provide doctors with a streamlined interface for managing their patient queue, conducting telemedicine sessions, and utilizing AI for clinical decision support.

Pages & Functionalities:

- Doctor Dashboard** (/dashboard/doctor-dashboard)
 - Purpose:* Displays the daily schedule, patient queue, and key statistics (e.g., total consultations completed).
 - Screenshot:* [Insert Doctor Dashboard Screenshot Here]
- Patient Management** (/dashboard/doctor-patients)
 - Purpose:* A comprehensive list of all assigned patients with search and filtering capabilities.
- Doctor Patient Profile** (/dashboard/doctor-patients/:id)
 - Purpose:* An in-depth view of a single patient's medical history, uploaded reports, and past prescriptions.
- Consultation Wizard & Video Room** (/dashboard/consultation/:id)
 - Purpose:* The core telemedicine feature. Doctors can join a video call with the patient while simultaneously viewing their history and taking notes.
- AI Analysis & Diagnosis** (/dashboard/ai-analysis, /dashboard/diagnosis)
 - Purpose:* Doctors can input clinical notes and utilize the AI to get secondary diagnostic suggestions and potential risk factors.
- Generate Report & Treatment Plans** (/dashboard/generate-report, /dashboard/doctor-treatment-plans)
 - Purpose:* Tools to create structured digital prescriptions, lab recommendations, and long-term care plans.

Dashboard 3: Admin Dashboard

Purpose:

To give system administrators full oversight of the platform's operations, user management, and financial health.

Pages & Functionalities:

- Admin Dashboard** (/dashboard/admin)
 - Purpose:* A bird's-eye view of system metrics, active users, total revenue, and system alerts.
 - Screenshot:* [Insert Admin Dashboard Screenshot Here]
- User & Doctor Management** (/dashboard/users, /dashboard/doctors)
 - Purpose:* Lists all registered patients and doctors. Admins can verify doctor credentials, suspend accounts, or reset passwords.
- Analytics Reports** (/dashboard/analytics)
 - Purpose:* Detailed graphs and charts showing platform growth, consultation volumes, and revenue trends.
- Manage Video Consultations** (/dashboard/admin-video-consults)
 - Purpose:* System-wide oversight of all active and scheduled telemedicine sessions. Admins can monitor room statuses.
- Financial Overview** (/dashboard/admin-payments)
 - Purpose:* Tracks all financial transactions, platform fees, and doctor payouts.

Page-Wise Workflow & Processes

1. The Consultation Workflow

- Step 1 (Patient):** The patient logs in, navigates to the *Appointments* page, selects a doctor, and books a time slot.
- Step 2 (System):** An invoice is generated in the *Billing* section. The patient completes the payment.
- Step 3 (Doctor):** The doctor sees the new appointment on their *Doctor Dashboard*.
- Step 4 (Execution):** At the scheduled time, both parties navigate to their respective *Video Room*. The doctor uses the *Consultation Wizard* to conduct the session.
- Step 5 (Post-Consultation):** The doctor generates a prescription and treatment plan. The patient immediately receives these in their *Reports* and *Treatment Plan* sections.

2. The AI Symptom Checker Workflow

- Step 1:** Patient navigates to *Symptom Checker*.
- Step 2:** They select or type their current symptoms and duration.
- Step 3:** The AI Service processes the data and returns potential conditions, severity levels, and whether a doctor's consultation is immediately required.
- Step 4:** The patient can click "Consult a Doctor" directly from the results page to seamlessly transition into booking an appointment.

3. The Admin Monitoring Workflow

- Step 1:** Admin logs into the system and opens the *Manage Video Consultations* page.
- Step 2:** The admin views a list of all scheduled, active, and completed rooms.
- Step 3:** Using the status filters, the admin isolates active rooms to ensure system stability and can click "Monitor" to resolve any technical disputes if necessary.

Conclusion

The AI Healthcare System leverages modern web technologies (React, Node.js, Express, MongoDB) and Artificial Intelligence to deliver a seamless, secure, and efficient healthcare experience. By defining clear boundaries and specialized tools for Patients, Doctors, and Admins, the platform ensures that users can focus on what matters most: delivering and receiving high-quality healthcare.

Note: This document provides the structural and textual foundation of the project. To finalize the professional PDF, actual system screenshots should be captured and inserted into the designated placeholder sections.

